

# DiceKriging vs RobustGaSP

Gu (2019)

## Examples

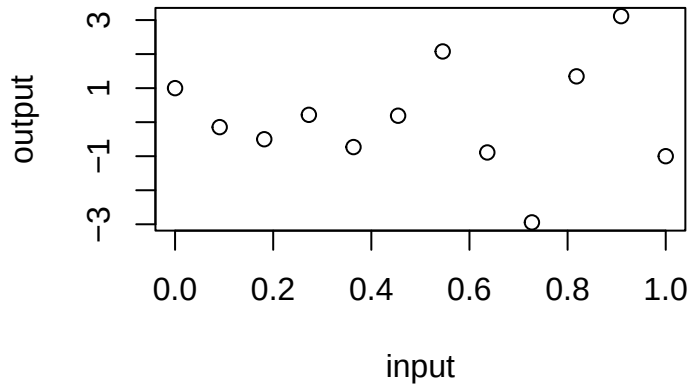
### The modified sine wave function

It is expected that, for a one-dimensional function, both packages will perform well with an adequate number of design points, so we start with the function called the modified sine wave discussed in Gu (2016). It has the form

$$y = 3 \sin(5\pi x) + \cos(7\pi x),$$

where  $x = [0, 10]$ . We first perform emulation based on 12 equally spaced design points on  $[0, 1]$ .

```
sinewave <- function(x) {  
  3*sin(5*pi*x)*x + cos(7*pi*x)  
}  
input <- as.matrix(seq(0, 1, 1/11))  
output <- sinewave(input)  
plot(output ~ input)
```



The GaSP model is attempted to be fitted by both the RobustGaSP and DiceKriging packages, but this time with a Gauss kernel, with the constant mean function. However, in the case of DiceKriging, an error is thrown (Error in chol.default(R) : the leading minor of order 8 is not positive) and so, to replicate the MLE approach of DiceKriging, RobustGaSP is used to fit the emulator (and then DiceKriging to make the predictions from the emulator object produced), setting the method argument equal to "mle". Whenever this is done, instead of using DiceKriging directly, it will be noted.

```
m <- rgasp(design=input, response=output,
           kernel_type = 'pow_exp',
           alpha = rep(2, ncol(input)),
           )
```

```
## The upper bounds of the range parameters are 0.7378104
## The initial values of range parameters are 0.01475621
## Start of the optimization 1 :
## The number of iterations is 8
## The value of the marginal posterior function is -19.61828
## Optimized range parameters are 0.04123854
## Optimized nugget parameter is 0
## Convergence: TRUE
## The initial values of range parameters are 0.01388889
## Start of the optimization 2 :
## The number of iterations is 8
## The value of the marginal posterior function is -19.61828
## Optimized range parameters are 0.04123854
## Optimized nugget parameter is 0
## Convergence: TRUE
```

```
m
```

```
##
## Call:
## rgasp(design = input, response = output, kernel_type = "pow_exp",
##       alpha = rep(2, ncol(input)))
## Mean parameters: 0.1440581
## Variance parameter: 2.543004
## Range parameters: 0.04123854
## Noise parameter: 0
```

```
dk <- rgasp(design=input, response=output,
            kernel_type = 'pow_exp',
            alpha = rep(2, ncol(input)),
            method = "mle"
            )
```

```
## The upper bounds of the range parameters are 0.7378104
## The initial values of range parameters are 0.01475621
## Start of the optimization 1 :
## The number of iterations is 1
## The value of the profile likelihood function is -19.97904
## Optimized range parameters are 0.01475621
## Optimized nugget parameter is 0
## Convergence: TRUE
## The initial values of range parameters are 0.01388889
## Start of the optimization 2 :
## The number of iterations is 1
## The value of the profile likelihood function is -19.97904
## Optimized range parameters are 0.01388889
## Optimized nugget parameter is 0
## Convergence: TRUE
```

```
dk
```

```
##  
## Call:  
## rgasp(design = input, response = output, method = "mle", kernel_type = "pow_exp",  
##       alpha = rep(2, ncol(input)))  
## Mean parameters:  0.1442577  
## Variance parameter:  2.327824  
## Range parameters:  0.01475621  
## Noise parameter:  0
```

The estimated range parameter is still found to be around 0.04 in the `RobustGaSP` package, but now 0.01 when using MLE. To see which estimate is better, we perform prediction on 100 test points, equally spaced in  $[0,1]$ .

```
testing_input <- as.matrix(seq(0, 1, 1/99))  
m.predict <- RobustGaSP::predict(m, testing_input)  
dk.predict <- DiceKriging::predict(dk, testing_input, type='UK', checkNames = F)
```

The emulation results are plotted in Figure 6. Note that the red curve from the `DiceKriging` package still degenerates to the fitted mean with spikes at the design points. This unsatisfying phenomenon, discussed in Gu et al. (2018), happens when the estimated covariance matrix is close to an identity matrix, i.e.,  $\hat{\mathbf{R}} \approx \mathbf{I}_n$ , or equivalently  $\hat{\gamma}$  tends to 0. Repeated runs of the `DiceKriging` package under different initializations yielded essentially the same results.

Changing from 5 to 12 and then to 32 observations has the following effects on the estimates. ‘`RobustGaSP`’ had to be used in place of ‘`DiceKriging`’ in the case of 12 observations (as above) and 32 observations. Note that between 25 and 30 observations, ‘`RobustGaSP`’ starts to produce odd results, as seen below.

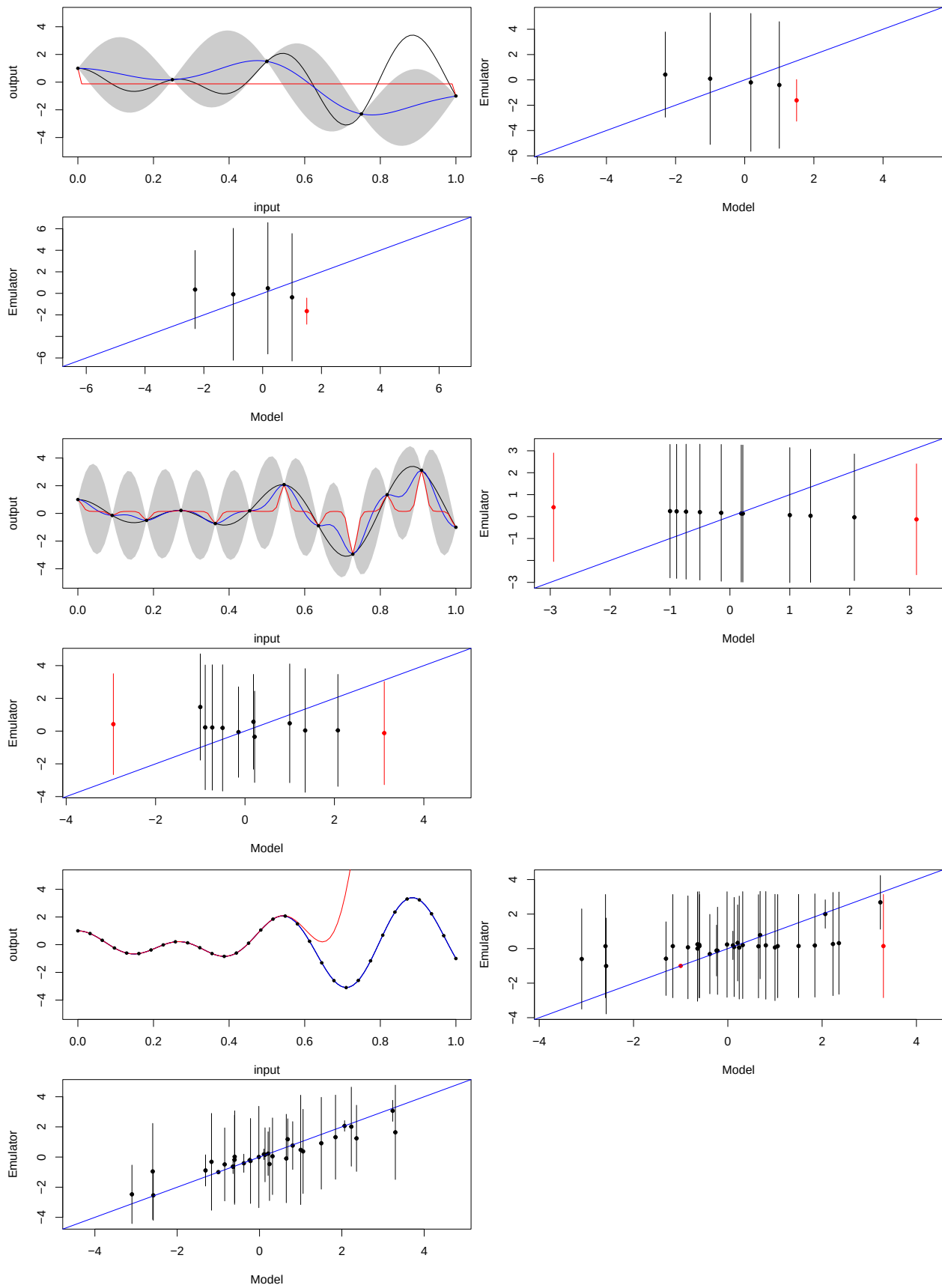


Table 1: Estimates table with 5 observations

| Package     | Kernel | Trend | Range | Variance |
|-------------|--------|-------|-------|----------|
| DiceKriging | Gauss  | -0.12 | 0     | 1.9      |
| RobustGaSP  | Gauss  | -0.11 | 0.17  | 2.49     |

Table 2: Estimates table with 12 observations

| Package     | Kernel | Trend | Range | Variance |
|-------------|--------|-------|-------|----------|
| DiceKriging | Gauss  | 0.14  | 0.01  | 2.33     |
| RobustGaSP  | Gauss  | 0.14  | 0.04  | 2.54     |

Table 3: Estimates table with 32 observations

| Package     | Kernel | Trend | Range | Variance |
|-------------|--------|-------|-------|----------|
| DiceKriging | Gauss  | 2.62  | 0.22  | 33.38    |
| RobustGaSP  | Gauss  | -0.1  | 0.16  | 3.25     |

## Verifying the estimates by-hand for both packages

To calculate the posterior predictions at the 100 testing points by-hand, we require the matrix  $H$  containing the inputs for the observations and well as matrix  $A$ , its inverse, and  $\mathbf{t}(x^*)$  for all 100 of the new inputs  $x^*$ .  $A$  and  $\mathbf{t}(x^*)$  are built using the estimate of the range parameter returned by each package. By-hand predictions match those returned by **DiceKriging** (Figure 2) and **RobustGaSP** (Figure 3).